AI & Modern Development

All 119 terms, A to Z. Each entry gives the idea in plain English first, then the glossary definition you'll be marked on, then the concepts an answer has to convey. The grey label on the right of each term is the section it comes from.

> **How to revise with this:** cover the two lower lines and read only the term. Say the definition out loud. Uncover and check whether you hit every concept listed under "must convey" — that, not the exact wording, is what counts. The tick list on the last page is for a final pass.

A

10 terms

Agent

Agents & Tools

A model given a goal and some tools, then left to run round a loop: decide, do it, look at the result, go again, until the job is done.

> An LLM given tools and a goal, running in a loop: decide → act → observe → repeat, until the task is done.

Must convey: an LLM with tools and a goal · runs in a loop until the task is done

Agentic Loop

Agents & Tools

That repeating think-then-act cycle. It's the thing separating an agent from a chatbot that answers once and stops.

> The repeating cycle of reasoning and action that distinguishes an agent from a single-shot chatbot.

Must convey: the repeating cycle of reasoning and action · what separates an agent from a chatbot

Alignment

Training

The whole effort to get a model doing what people actually intend and value, rather than something technically correct but unwanted.

> The broad effort to make a model's behaviour match human intent and values.

Must convey: making behaviour match human intent and values

API (Application Programming Interface)

APIs & Building

An agreed way for one program to ask another program for something.

> A defined way for one program to request something from another.

Must convey: a defined way for one program to request from another

API Key

APIs & Building

The credential proving a request really came from you. Treat it exactly like a password, and never let it get committed.

> A credential authenticating your requests. Treat it like a password: never commit it to a repo.

Must convey: a credential that authenticates your requests · treat it like a password — never commit it

Architecture

Foundations

The shape of a model: how its layers are wired together. Two models can share the same shape and have learned completely different numbers.

> The structural blueprint of a model (e.g. Transformer, CNN, RNN). Two models can share an architecture but have completely different weights.

Must convey: the structural blueprint of a model · separate from the weights

Artificial Intelligence (AI)

Foundations

Any software doing a job you'd normally need a person's judgement for. It's a label for a whole field, not one particular gadget.

> Software that performs tasks normally requiring human judgment — recognizing, predicting, generating, deciding. An umbrella term, not a specific technology.

Must convey: tasks needing human judgment · an umbrella term, not one technology

Attention
How Language Models Work

The mechanism that lets the model decide which earlier words matter right now. When a sentence looks at itself this way, it's called self-attention.

> The mechanism letting the model weigh which earlier tokens matter when producing the next one. “Self-attention” = the sequence attending to itself.

`Must convey: weighs which earlier tokens matter · self-attention = the sequence attends to itself`

Automation Bias
Evaluation, Risk & Ethics

People's habit of trusting the machine too readily. The most common real AI failure isn't the model being wrong, it's nobody checking.

> The human tendency to over-trust machine output. The most common real-world AI failure is a person not checking.

`Must convey: people over-trust machine output · the real failure is a person not checking`

Autoregressive
How Language Models Work

Writing one piece at a time, where each piece depends on everything already written. That's why answers appear word by word.

> Generating one token at a time, each conditioned on all previous ones. Why responses stream in word by word.

`Must convey: one token at a time · each conditioned on all the previous ones`

B
5 terms

Backpropagation
Training

Working backwards from the mistake to work out how much each individual number was to blame. That blame is what gets used to fix things.

> The algorithm that computes how much each weight contributed to the error, working backward from output to input. It produces the gradients.

`Must convey: how much each weight contributed to the error · works backward and produces the gradients`

Barge-in
Voice Agents

Letting the person cut in while the agent is still talking, and having it stop promptly. Without this, conversation feels wrong.

> Letting the user interrupt mid-response and having the agent stop talking. Essential for natural conversation.

`Must convey: the user interrupts mid-response · and the agent stops talking`

Benchmark
Evaluation, Risk & Ethics

A public standard test used to compare models. Useful for a rough sense of direction, but widely gamed.

> A standardized public eval used to compare models. Useful directionally; often gamed.

`Must convey: a standardized public eval for comparing models · directional, and often gamed`

Bias
Evaluation, Risk & Ethics

A consistent unfair lean in what comes out, inherited from the training data or from choices made while building it.

> Systematic skew in outputs inherited from training data or design choices.

`Must convey: systematic skew in outputs · inherited from training data or design choices`

Branch
Version Control & GitHub

A parallel version of the project, so you can work without touching the main one.

> A parallel line of development. Lets you work without disturbing the main version.

`Must convey: a parallel line of development · work without disturbing main`

C
8 terms

Chain of Thought (CoT)
Prompting

Telling the model to work through it step by step before answering. It gets noticeably better at anything with several stages.

> Prompting the model to reason step by step before answering. Improves accuracy on multi-step problems.

`Must convey: reason step by step before answering · improves accuracy on multi-step problems`

Chunking

Retrieval & Knowledge

Cutting documents into passages small enough to fetch and small enough to fit. How big you cut them, and how much they overlap, quietly decides how good your results are.

> Splitting documents into passages small enough to retrieve and fit in context. Chunk size and overlap materially affect RAG quality.

Must convey: splitting documents into passages · small enough to retrieve and fit in context

CI/CD

Version Control & GitHub

Continuous integration and continuous deployment: testing and shipping your changes automatically instead of by hand.

> Continuous Integration / Continuous Deployment. Automatically testing and shipping code changes.

Must convey: continuous integration and continuous deployment · automatic testing and shipping of changes

Clone

Version Control & GitHub

Downloading your own full copy of a remote project.

> Downloading a full copy of a remote repo to your machine.

Must convey: downloading a full copy of a remote repo

Commit

Version Control & GitHub

A saved snapshot of your changes, with a note explaining them. It's the basic unit of the history.

> A saved snapshot of changes with a message explaining them. The atomic unit of Git history.

Must convey: a saved snapshot of changes · with a message; the atomic unit of history

Container / Docker

APIs & Building

Boxing up an application with everything it needs, so it runs the same way on any machine.

> Packaging an application with all its dependencies so it runs identically everywhere.

Must convey: packaging an app with all its dependencies · so it runs identically everywhere

Context

Prompting

Everything the model can currently see: the standing instructions, the conversation so far, any documents fetched in, and any tool results.

> Everything currently in the model's window — system prompt, history, retrieved documents, tool results.

Must convey: everything currently in the model's window · system prompt, history, documents, tool results

Context Window

How Language Models Work

How much the model can hold in its head at once, counting your prompt and its reply together. Fill it up and the oldest material falls out the back.

> The maximum number of tokens a model can consider at once — prompt plus response. Exceed it and the earliest content falls out of view.

Must convey: the maximum tokens it can consider at once · prompt plus response; overflow falls out

D

4 terms

Data Residency

Evaluation, Risk & Ethics

Which country your data physically sits in, and therefore whose laws apply to it.

> Where data is physically stored and which laws therefore apply to it.

Must convey: where data is physically stored · and which laws apply to it

Deep Learning

Foundations

Machine learning done with neural networks stacked many layers deep. Essentially all modern AI is this.

> ML using neural networks with many layers. Nearly all modern AI (image, speech, language) is deep learning.

Must convey: uses neural networks · with many layers

Diarization

Voice Agents

Working out who said which part when several people are talking.

> Identifying who spoke when in multi-speaker audio.

Must convey: identifying who spoke when · in multi-speaker audio

Distillation

Training

Have a small model copy a big one's answers. You keep most of the ability at a fraction of the size.

> Training a small model to imitate a large one, capturing much of the capability at a fraction of the size.

Must convey: a small model imitates a large one · keeps much capability at a fraction of the size

E

6 terms

Embedding

How Language Models Work

A list of numbers that captures what something means, arranged so that things with similar meanings get similar numbers.

> A list of numbers (a vector) representing meaning. Similar meanings sit close together in that numeric space. Foundation of search, RAG, and recommendation.

Must convey: a vector of numbers · representing meaning — similar things sit close

Endpoint

APIs & Building

One specific web address an API offers, for one specific job.

> A specific URL an API exposes for a specific operation.

Must convey: a specific URL for a specific operation

Environment Variable

APIs & Building

A setting kept outside your code, which is where your keys belong.

> A configuration value stored outside your code — the correct home for API keys.

Must convey: a config value stored outside your code · the right home for API keys

Epoch / Batch

Training

An epoch is one complete lap through all the training data. A batch is the handful of examples it looks at before each nudge.

> An epoch is one full pass through the training data. A batch is the chunk of examples processed before each weight update.

Must convey: epoch = one full pass through the data · batch = the chunk before each update

Eval

Evaluation, Risk & Ethics

A proper structured test of how well the thing performs at a defined job. If you're not measuring it, you can't tell whether you're improving it.

> A structured test measuring model or system performance on a defined task. If you can't measure it, you can't improve it.

Must convey: a structured test measuring performance on a task · you can't improve what you don't measure

Explainability

Evaluation, Risk & Ethics

How far you can actually look inside and follow why a model decided something. For big models, not very far.

> The degree to which a model's reasoning can be inspected and understood. Generally poor for large models.

Must convey: how far a model's reasoning can be inspected · generally poor for large models

F

2 terms

Fine-tuning

Training

Taking a model that's already trained and training it a bit more on your narrower material, so it gets good at your particular job. Far cheaper than starting over.

> Further training of an existing model on a narrower dataset to specialize it. Vastly cheaper than pre-training.

Must convey: further training of an existing model · on a narrower dataset, to specialize it

Fork

Version Control & GitHub

Taking your own copy of somebody else's project, under your account.

> Making your own copy of someone else's repo under your account.

Must convey: your own copy of someone else's repo

G

8 terms

Git

Version Control & GitHub

The tool that records every change to your files, who made it, and when. It runs on your own machine.

> The version control system. Tracks every change to your files, who made it, and when. Runs locally.

`Must convey: a version control system tracking every change · runs locally`

GitHub

Version Control & GitHub

A website that hosts Git projects and adds the teamwork parts on top. Git is the tool, GitHub is the place; they are not the same thing.

> A hosting platform for Git repositories, plus collaboration tools on top. Git ≠ GitHub.

`Must convey: hosts Git repositories · adds collaboration tools; not the same as Git`

GitHub Actions

Version Control & GitHub

Jobs that run automatically when something happens in the project. Run the tests on every push; deploy on every merge.

> Automation that runs on repo events: test on every push, deploy on every merge.

`Must convey: automation that runs on repo events · e.g. test on push, deploy on merge`

.gitignore

Version Control & GitHub

A list of things Git should never record: secrets, build output, and downloaded dependencies.

> A file listing what Git should never track — secrets, build output, dependencies.

`Must convey: lists what Git should never track · secrets, build output, dependencies`

Gradient

Training

For one number inside the model: which direction to nudge it, and how hard, to make the mistake smaller.

> The direction and magnitude of change that would reduce the loss for a given weight.

`Must convey: direction and size of a change · that would reduce the loss`

Gradient Descent

Training

Actually doing the nudging, over and over. This repeated nudging is the whole of what people call learning.

> Repeatedly nudging weights in the direction the gradients indicate. The actual "learning" step.

`Must convey: nudging weights along the gradients · repeatedly — this is the learning`

Grounding

Retrieval & Knowledge

Anchoring what the model says to sources somebody can actually go and check.

> Tying a model's output to verifiable sources.

`Must convey: tying output to verifiable sources`

Guardrails

Agents & Tools

The limits on what an agent is allowed to do: which tools, how much it may spend, and what needs a human to say yes first.

> Constraints on what an agent may do — allowed tools, spending caps, human approval for risky actions.

`Must convey: constraints on what an agent may do · allowed tools, spend caps, approval for risk`

H

2 terms

Hallucination

Evaluation, Risk & Ethics

The model stating something completely wrong with total confidence. It isn't lying, because it has no idea what's true, only what sounds likely. Check anything that matters.

> Confidently stated output that is factually wrong. Not lying — the model has no concept of truth, only likelihood. Always verify consequential claims.

`Must convey: confidently stated output that is factually wrong · not lying — it tracks likelihood, not truth`

Human in the Loop (HITL)

Agents & Tools

A deliberate stopping point where a person has to approve before the agent does anything that really matters.

> A checkpoint requiring human approval before an agent takes a consequential action.

Must convey: a human approval checkpoint · before a consequential action

I

2 terms

Inference

Inference & Running Models

Running the finished model to get an answer out of it. Every message you send is one of these.

> Running a trained model to get an output. Every chat message you send is an inference request.

Must convey: running a trained model to get output · every request you send is one

Issue

Version Control & GitHub

A logged task, bug or feature request.

> A tracked task, bug, or feature request.

Must convey: a tracked task, bug or feature request

J

2 terms

Jailbreak

Evaluation, Risk & Ethics

A prompt written specifically to get around a model's safety behaviour.

> A prompt crafted to bypass a model's safety behaviour.

Must convey: a prompt crafted to bypass safety behaviour

JSON

APIs & Building

The common text format for passing structured data around. Names, values, and things nested inside other things.

> The standard text format for structured data exchange. Keys, values, nesting.

Must convey: the standard text format for structured data · keys, values, nesting

K

2 terms

Knowledge Cutoff

Retrieval & Knowledge

The date the model's training stopped. It knows nothing after that unless you hand the information to it.

> The date after which a model has no training knowledge. Anything later must be supplied via search or retrieval.

Must convey: the date after which it knows nothing · later facts must be supplied to it

KV Cache

Inference & Running Models

Keeping the working-out from earlier words so it doesn't have to redo it every time. Faster, at the cost of memory.

> Stored intermediate values from earlier tokens so they don't need recomputing. Speeds up generation, consumes memory.

Must convey: stores intermediate values from earlier tokens · avoids recomputing; costs memory

L

8 terms

Latency

Inference & Running Models

How long you wait between asking and getting an answer. In voice work this is the number that decides whether it feels alive.

> Time from request to response. In voice work, the number that makes or breaks the experience.

Must convey: time from request to response · decisive in voice work

Latency Budget

Voice Agents

The total round-trip time you're allowed, with every stage spending out of the same pot. Under about eight hundred milliseconds feels like talking; past a second and a half feels broken.

> The total allowable round-trip time. Under ~800 ms feels conversational; past ~1.5 s feels broken. Every stage spends from this budget.

Must convey: the total allowable round-trip time · thresholds where it feels natural or broken · every stage spends from it

Learning Rate

Training

How big each nudge is. Too big and it thrashes about and never settles. Too small and you'll be waiting forever.

> How big each nudge is. Too high, training diverges; too low, it never finishes.

Must convey: how big each update step is · too high diverges, too low never finishes

LLM (Large Language Model)

How Language Models Work

A very large neural network whose only real skill is guessing the next chunk of text. Do that well enough and conversation, code and reasoning come out of it.

> A large neural network trained to predict the next token in a sequence. That single objective, at scale, produces reasoning, translation, coding, and conversation.

Must convey: a large neural network · trained to predict the next token

Local Inference

Inference & Running Models

Running the model on your own machine instead of calling somebody's cloud. Private, and free per use, but limited by your graphics card.

> Running a model on your own hardware rather than calling a cloud API. Better privacy and no per-token cost; limited by your GPU.

Must convey: running on your own hardware, not a cloud API · private and free, but GPU-limited

Localhost

APIs & Building

Your own machine, talking to itself as though it were a server. It's where you try things before putting them live.

> Your own machine, addressed as a server. Where you test before deploying.

Must convey: your own machine addressed as a server · where you test before deploying

LoRA (Low-Rank Adaptation)

Training

A cheap way to fine-tune: leave the big model alone and train a small bolt-on piece instead. Quick, and you can swap the piece out.

> A fine-tuning method that trains a small add-on layer instead of the whole model. Fast, cheap, and swappable.

Must convey: trains a small add-on instead of the whole model · cheap, fast and swappable

Loss Function

Training

The score for how wrong the model's guess was. Training is simply the business of driving that score down.

> The formula measuring how wrong a prediction was. Training = making this number go down.

Must convey: measures how wrong a prediction was · training pushes it down

M

9 terms

Machine Learning (ML)

Foundations

Instead of you writing the rules, you show the computer thousands of examples and it works the rules out for itself.

> A subset of AI where a system learns patterns from data instead of following hand-written rules.

Must convey: learns patterns from data · instead of hand-written rules

main

Version Control & GitHub

The usual name for a project's primary branch.

> The conventional name for the primary branch.

Must convey: the conventional name for the primary branch

MCP (Model Context Protocol)

Agents & Tools

A shared standard for plugging models into outside tools and data, so every connection isn't a bespoke job.

> An open standard for connecting models to external tools and data sources — a universal adapter so every integration isn't bespoke.

Must convey: an open standard connecting models to tools and data · a universal adapter instead of bespoke work

Merge

Version Control & GitHub

Bringing one branch's changes into another.

> Combining one branch's changes into another.

Must convey: combining one branch's changes into another

Merge Conflict

Version Control & GitHub

Two branches edited the same lines and Git won't guess which you meant. You have to sort it out by hand.

> When two branches change the same lines and Git can't decide which wins. You resolve it by hand.

Must convey: two branches changed the same lines · Git can't decide; you resolve it by hand

Mixture of Experts (MoE)

How Language Models Work

A design where only a slice of the model wakes up for each token. You get big-model quality without paying big-model running costs.

> An architecture where only a fraction of the model's parameters activate per token. Gives large-model quality at smaller-model inference cost.

Must convey: only some parameters activate per token · large-model quality at lower inference cost

Model

Foundations

The finished trained thing: a big file of learned numbers, plus the design that knows how to use them. When you download a model, that file is what you're getting.

> The trained artifact — the file(s) of learned numbers plus the architecture that uses them. “Downloading a model” means downloading those weights.

Must convey: the trained artifact / learned weights · plus the architecture that uses them

Multi-Agent System

Agents & Tools

Several specialised agents splitting the work, usually with one in charge handing tasks out.

> Several specialized agents dividing a task, often with a coordinator delegating to workers.

Must convey: several specialized agents divide a task · often a coordinator delegating to workers

Multimodal

How Language Models Work

A model that takes more than just text: pictures, sound or video as well.

> A model that handles more than one input type — text plus images, audio, or video.

Must convey: handles more than one input type · text plus images, audio or video

N

2 terms

Neural Network

Foundations

Layers of tiny units, each adding up the numbers coming in and passing a signal on. Loosely like brain cells, but really it's just an enormous amount of multiplication.

> A model made of layers of interconnected “neurons,” each applying a weighted sum plus a nonlinearity. Loosely brain-inspired; mathematically just matrix multiplication at scale.

Must convey: layers of connected neurons · weighted sum plus a nonlinearity

Next-Token Prediction

How Language Models Work

The model scores every possible next chunk of text, one gets picked, it's stuck on the end, and then the whole thing runs again.

> The core operation. The model outputs a probability distribution over the whole vocabulary, then one token is chosen and appended, and the process repeats.

Must convey: a probability distribution over the vocabulary · one token is chosen, appended, and it repeats

O

3 terms

Open Source

Version Control & GitHub

Code published with a licence letting other people use it, change it and pass it on. The exact conditions differ by licence, so read it.

> Code published under a license permitting use, modification, and redistribution. Terms vary by license — read them.

Must convey: published under a permissive license · allows use, modification, redistribution — terms vary

Orchestration

Agents & Tools

Wiring several models, agents or steps together into one system that works.

> Coordinating multiple models, agents, or steps into a working system.

Must convey: coordinating multiple models, agents or steps · into one working system

Overfitting

Training

When the model has effectively memorised the practice papers. Brilliant on those, useless on anything new.

> The model memorizes training data instead of generalizing. Performs great on training data, poorly on new data.

Must convey: memorizes the training data · fails on new, unseen data

P

9 terms

Phoneme

Voice Agents

The smallest unit of speech sound. You reach for these to force the right pronunciation of names and technical words.

> The smallest unit of speech sound. Used to fix mispronunciations of names and technical terms.

Must convey: the smallest unit of speech sound · used to fix mispronunciations

PII (Personally Identifiable Information)

Evaluation, Risk & Ethics

Information that identifies a real person. It governs what you're allowed to send off to somebody else's API, and the rules depend on where you are.

> Data identifying an individual. Governs what you may send to a third-party API — check your jurisdiction's privacy law.

Must convey: data identifying an individual · governs what you may send third parties; privacy law

Pre-training

Training

The enormous first training run over a mountain of general text. This is where the broad ability comes from, and it costs millions.

> The initial, massive training run on broad data. Where general capability comes from. Costs millions of dollars.

Must convey: the initial massive run on broad data · source of general capability; very expensive

Prompt

Prompting

The text you give the model.

> The input text given to a model.

Must convey: the input text you give the model

Prompt Engineering

Prompting

The craft of wording a request so you reliably get what you want. Less mystical than it sounds: be specific, show an example, say what format you want.

> The practice of structuring inputs to get reliable outputs. Less mystical than it sounds: be specific, give examples, state the format.

Must convey: structuring inputs to get reliable outputs · be specific, give examples, state the format

Prompt Injection

Prompting

An attack where instructions are smuggled inside content the model reads, such as a web page or an email, and it obeys them instead of you. Still unsolved, and serious once agents can act on it.

> An attack where malicious instructions hidden in data (a webpage, an email, a document) hijack the model's behaviour. A live, unsolved security problem for agents.

Must convey: malicious instructions hidden in data · hijacks the model's behaviour · an unsolved security problem for agents

Prosody

Voice Agents

The rhythm, stress and tune of speech. It's what makes a synthetic voice sound human rather than robotic.

> The rhythm, stress, and intonation of speech — what makes synthesized voice sound human or robotic.

Must convey: the rhythm, stress and intonation of speech · what makes a voice sound human or robotic

Pull Request (PR)

Version Control & GitHub

You proposing that your branch be merged in, opened up so others can review it and argue about it first. It's the centre of how teams work.

> A proposal to merge your branch, opened for review and discussion. The heart of collaborative workflow.

Must convey: a proposal to merge your branch · opened for review and discussion

Push / Pull

Version Control & GitHub

Push sends your commits up to the remote. Pull brings other people's commits down to you.

> Push = send your local commits to the remote. Pull = fetch and apply the remote's commits locally.

Must convey: push sends local commits to the remote · pull fetches and applies remote commits locally

Q

1 term

Quantization

Inference & Running Models

Squashing the model's numbers into fewer bits so it fits on smaller hardware. You lose a sliver of accuracy and save a mountain of memory.

> Compressing weights to lower precision (e.g. 16-bit → 4-bit) so a model fits smaller hardware. Trades a little accuracy for a lot of memory.

Must convey: compressing weights to lower precision · so it fits smaller hardware · trades a little accuracy

R

6 terms

RAG (Retrieval-Augmented Generation)

Retrieval & Knowledge

Go and fetch the relevant documents, paste them into the prompt, and make the model answer from those rather than from memory. This is the standard way to point AI at your own material.

> Fetching relevant documents and inserting them into the prompt so the model answers from real sources instead of memory. The standard pattern for grounding AI in your own data.

Must convey: fetch relevant documents into the prompt · so it answers from real sources, not memory

Rate Limit

APIs & Building

A ceiling on how many requests you may make in a given time. Go over it and you start getting errors.

> A cap on requests per time period. Exceed it and you get errors.

Must convey: a cap on requests per time period · exceed it and you get errors

README.md

Version Control & GitHub

The front page of a project: what it is, how to install it, how to use it. Write it first, not last.

> The front page of a repo. What the project is, how to install it, how to use it. Write it first, not last.

Must convey: the front page of a repo · what it is, how to install, how to use

Red Teaming

Evaluation, Risk & Ethics

Deliberately attacking your own system to find out how it breaks, before somebody less friendly does.

> Deliberately attacking your own system to find failure modes before someone else does.

Must convey: deliberately attacking your own system · to find failures before someone else does

Repository (Repo)

Version Control & GitHub

A project folder that Git is watching, complete with its entire history.

> A project folder tracked by Git, including its full history.

Must convey: a project folder tracked by Git · including its full history

RLHF (Reinforcement Learning from Human Feedback)

Training

Show people two answers, ask which they prefer, and train the model towards the preferred one. This is the step that turns a text predictor into something that behaves like an assistant.

> Training a model against human preference ratings to make it more helpful and less harmful. What turns a raw text predictor into an assistant.

Must convey: trained against human preference ratings · makes it a helpful, less harmful assistant

S

9 terms

Sampling

Inference & Running Models

The actual act of picking the next word from all the scored candidates.

> The act of choosing the next token from the probability distribution.

Must convey: choosing the next token from the distribution

SDK

APIs & Building

A library in your own language that wraps an API so you write far less plumbing.

> A language-specific library wrapping an API so you write less boilerplate.

Must convey: a language-specific library wrapping an API · so you write less boilerplate

Semantic Search

Retrieval & Knowledge

Searching by what you meant, rather than by the exact words you typed.

> Search by meaning rather than keyword match.

Must convey: search by meaning, not keyword

Speech-to-Speech

Voice Agents

One model taking audio straight in and putting audio straight out, skipping the written step in the middle. Faster, and it keeps the feeling in the voice, but much harder to see inside.

> A single model handling audio in and audio out, skipping the text round trip. Lower latency, preserves tone and emotion, harder to inspect.

Must convey: one model, audio in and audio out · lower latency and keeps tone, but harder to inspect

SSML

Voice Agents

Small tags you wrap around text to tell the speech engine where to pause, what to stress, and how to pronounce things.

> A markup language for controlling TTS output: pauses, emphasis, pronunciation, speed.

Must convey: a markup language for controlling TTS output · pauses, emphasis, pronunciation, speed

State / Memory

Agents & Tools

Whatever the agent carries forward between steps, or between sessions. Models remember nothing by default, so this has to be deliberately built.

> Information an agent carries between steps or sessions. Models are stateless by default; memory must be engineered.

Must convey: information carried between steps or sessions · models are stateless by default — memory is built

Streaming
Inference & Running Models

Sending words out as they're produced, instead of making you wait for the whole finished answer.

> Sending tokens to the user as they're generated rather than waiting for the full response.

Must convey: sending tokens as they are generated · rather than waiting for the whole response

STT / ASR (Speech-to-Text / Automatic Speech Recognition)
Voice Agents

Turning spoken audio into written words. Whisper is the well-known open one.

> Converting spoken audio into text. Whisper is the common open model.

Must convey: converting spoken audio into text

System Prompt
Prompting

The standing instructions sitting behind the conversation, setting the model's role, its rules and how it should behave. You don't see it in the chat.

> Instructions setting the model's role, constraints, and behaviour, separate from the user's message.

Must convey: sets the model's role, constraints and behaviour · separate from the user's message

T
11 terms

Temperature
Inference & Running Models

The randomness dial. At zero it says the same safe thing every time. Turn it up and it gets inventive, then unhinged.

> Controls randomness in token selection. 0 = deterministic and repetitive; high = creative and erratic.

Must convey: controls randomness in token selection · 0 is deterministic, high is erratic

Throughput
Inference & Running Models

How many tokens come out per second.

> Tokens produced per second.

Must convey: tokens produced per second

Token
How Language Models Work

The bite-sized piece of text a model actually reads, roughly three quarters of a word. It never sees individual letters, which is why it miscounts them.

> The unit an LLM reads and writes. Roughly ¾ of a word in English — “unbelievable” might be un + believ + able. Models see tokens, never letters, which is why they struggle to count characters.

Must convey: the unit a model reads and writes · roughly part of a word, not a letter

Tokenizer
How Language Models Work

The piece of software that chops your text into those pieces and numbers them, then puts them back together afterwards.

> The component that converts text into token IDs and back. Different model families use different tokenizers.

Must convey: converts text into token IDs · and back again

Tool Use / Function Calling
Agents & Tools

The model asks, in a strict format, for a particular function to be run. The answer comes back to it and it carries on from there.

> The model outputs a structured request to run a function (search, query a database, send an email), receives the result, and continues.

Must convey: outputs a structured request to run a function · gets the result back and continues

Top-p / Top-k
Inference & Running Models

Other dials doing a similar job: they cut down the shortlist of words the model is allowed to pick from.

> Alternative sampling controls limiting the pool of candidate next tokens.

Must convey: sampling controls · they limit the pool of candidate tokens

Training vs. Inference

Foundations

Training is the school years: slow, expensive, done once. Inference is the day job: using the trained model to answer you, over and over.

> Training = teaching the model (expensive, done once). Inference = using the trained model to produce output (cheap-ish, done constantly).

`Must convey: training teaches the model — expensive, done once · inference is using it — cheap, done constantly`

Transformer

Foundations

The particular design nearly every modern language model uses. Its clever trick is attention. It arrived in twenty seventeen.

> The architecture behind virtually all modern language models. Its key innovation is attention. Introduced in 2017.

`Must convey: the architecture behind modern language models · its key innovation is attention`

TTFT (Time To First Token)

Inference & Running Models

How long before the very first word appears. This matters more to how fast it feels than the total time does.

> How long until the first token appears. Perceived responsiveness depends on this more than total time.

`Must convey: how long until the first token appears · drives perceived responsiveness`

TTS (Text-to-Speech)

Voice Agents

Turning written words into spoken audio.

> Converting text into spoken audio.

`Must convey: converting text into spoken audio`

Turn Detection / Endpointing

Voice Agents

Deciding the person has finished speaking, so the agent can reply. Get it wrong and it either talks over them or sits there seeming slow.

> Deciding when the user has finished their turn so the agent can respond. Get it wrong and the agent either interrupts or feels sluggish.

`Must convey: deciding the user has finished their turn · so the agent can respond; errors interrupt or lag`

V

6 terms

VAD (Voice Activity Detection)

Voice Agents

Working out whether somebody is actually talking right now, or whether that's just the room.

> Detecting when someone is actually speaking versus background noise.

`Must convey: detecting when someone is actually speaking · as opposed to background noise`

Vector

How Language Models Work

Just an ordered list of numbers. Embeddings are vectors, and so is nearly everything inside a model.

> An ordered list of numbers. An embedding is a vector; so is nearly everything inside a model.

`Must convey: an ordered list of numbers · embeddings are vectors — they're everywhere inside a model`

Vector Database

Retrieval & Knowledge

A store for embeddings that can quickly find the entries closest in meaning to whatever you asked.

> A database that stores embeddings and finds the nearest matches by meaning. Examples: Pinecone, Chroma, Qdrant, pgvector.

`Must convey: stores embeddings · finds nearest matches by meaning`

Voice Cloning

Voice Agents

Making speech in one particular person's voice from a short recording of them. Powerful, and an ethical minefield: consent is the whole issue.

> Generating speech in a specific person's voice from a short sample. Powerful and ethically loaded — consent matters.

`Must convey: speech in a specific person's voice from a sample · consent matters — ethically loaded`

Voice Pipeline
Voice Agents

The classic three-step build: speech to text, then the model, then text to speech. Easy to inspect, but every step adds delay.

> The classic three-stage design: STT → LLM → TTS. Modular and debuggable, but latency accumulates at each stage.

`Must convey: three stages: STT then LLM then TTS · modular but latency accumulates`

VRAM
Inference & Running Models

The memory on your graphics card. In practice it's the hard ceiling on how big a model you can run at home.

> Video memory on a GPU. The practical ceiling on what model you can run locally.

`Must convey: video memory on a GPU · the ceiling on what you can run locally`

W
3 terms

Webhook
APIs & Building

An API call in reverse: instead of you asking, the service calls you when something happens.

> A reverse API call: the service notifies you when something happens.

`Must convey: a reverse API call · the service notifies you when something happens`

Weights / Parameters
Foundations

The numbers the model learned during training. Seventy billion parameters means seventy billion of those numbers. More of them usually means cleverer, and hungrier for memory.

> The learned numbers inside a model. “70B parameters” = 70 billion of them. More parameters generally means more capability and more memory required.

`Must convey: the learned numbers inside a model · count tracks capability and memory`

WER (Word Error Rate)
Voice Agents

The standard score for how badly speech recognition mangled the words. Lower is better, and accents, jargon and background noise all push it up.

> The standard accuracy measure for speech recognition. Lower is better. Accents, jargon, and background noise all raise it.

`Must convey: the accuracy measure for speech recognition · lower is better; noise and accents raise it`

Z
1 term

Zero-shot / Few-shot
Prompting

Zero-shot is asking cold, with no examples. Few-shot is showing it two or three worked examples first.

> Asking with no examples vs. including a few worked examples in the prompt.

`Must convey: zero-shot = no examples · few-shot = a few examples in the prompt`

✓ FINAL PASS — CAN YOU DEFINE EACH ONE?

119 TERMS

- ☐ Agent
- ☐ Agentic Loop
- ☐ Alignment
- ☐ API (Application Programming Interface)
- ☐ API Key
- ☐ Architecture
- ☐ Artificial Intelligence (AI)
- ☐ Attention
- ☐ Automation Bias
- ☐ Autoregressive
- ☐ Backpropagation
- ☐ Barge-in
- ☐ Benchmark
- ☐ Bias
- ☐ Branch
- ☐ Chain of Thought (CoT)
- ☐ Chunking
- ☐ CI/CD
- ☐ Clone
- ☐ Commit
- ☐ Container / Docker
- ☐ Context
- ☐ Context Window
- ☐ Data Residency
- ☐ Deep Learning
- ☐ Diarization
- ☐ Distillation
- ☐ Embedding
- ☐ Endpoint
- ☐ Environment Variable
- ☐ Epoch / Batch
- ☐ Eval
- ☐ Explainability
- ☐ Fine-tuning
- ☐ Fork
- ☐ Git
- ☐ GitHub
- ☐ GitHub Actions
- ☐ .gitignore
- ☐ Gradient
- ☐ Gradient Descent
- ☐ Grounding
- ☐ Guardrails
- ☐ Hallucination
- ☐ Human in the Loop (HITL)
- ☐ Inference
- ☐ Issue
- ☐ Jailbreak
- ☐ JSON
- ☐ Knowledge Cutoff
- ☐ KV Cache
- ☐ Latency
- ☐ Latency Budget
- ☐ Learning Rate
- ☐ LLM (Large Language Model)
- ☐ Local Inference
- ☐ Localhost
- ☐ LoRA (Low-Rank Adaptation)
- ☐ Loss Function
- ☐ Machine Learning (ML)
- ☐ main
- ☐ MCP (Model Context Protocol)
- ☐ Merge
- ☐ Merge Conflict
- ☐ Mixture of Experts (MoE)
- ☐ Model
- ☐ Multi-Agent System
- ☐ Multimodal
- ☐ Neural Network
- ☐ Next-Token Prediction
- ☐ Open Source
- ☐ Orchestration
- ☐ Overfitting
- ☐ Phoneme
- ☐ PII (Personally Identifiable Information)
- ☐ Pre-training
- ☐ Prompt
- ☐ Prompt Engineering
- ☐ Prompt Injection
- ☐ Prosody
- ☐ Pull Request (PR)
- ☐ Push / Pull
- ☐ Quantization
- ☐ RAG (Retrieval-Augmented Generation)
- ☐ Rate Limit
- ☐ README.md
- ☐ Red Teaming
- ☐ Repository (Repo)
- ☐ RLHF (Reinforcement Learning from Human Feedback)
- ☐ Sampling
- ☐ SDK
- ☐ Semantic Search
- ☐ Speech-to-Speech
- ☐ SSML
- ☐ State / Memory
- ☐ Streaming
- ☐ STT / ASR (Speech-to-Text / Automatic Speech Recognition)
- ☐ System Prompt
- ☐ Temperature
- ☐ Throughput
- ☐ Token
- ☐ Tokenizer
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